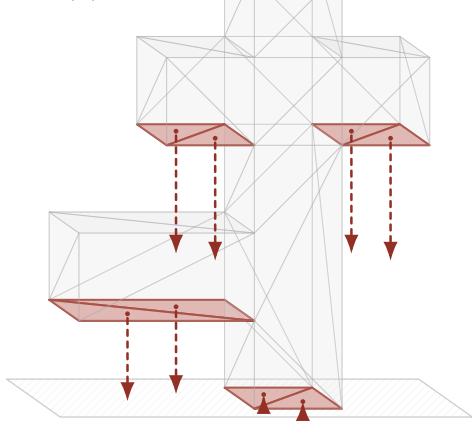


FTree4x source-face sampling

$A_i O_i(n) > 0$ faces selected for ray casting



red faces: source pool; dashed arrows: rays along $-n$

Ray budget on FTree4x

$$K_{ray} = \text{clip}(0.02N_f, 500, 3000)$$

$$N_f = 60, \quad 0.02N_f = 1.2$$

$$K_{ray} = \text{clip}(1.2, 500, 3000) = 500$$

weighted source pool: $A_i O_i(n) > 0$

$$N_{src} = 8 < K_{ray}$$

selected ray sources: 8 faces



small meshes: minimum 500 protects stability

medium meshes: sample about 2% of faces

large meshes: cap 3000 keeps runtime bounded

For this small didactic mesh, the lower bound exceeds the overhang source pool, so no subsampling occurs: every weighted source face is ray-cast.